

# Breaking the Reversal Curse in Autoregressive Language Models via Identity Bridge

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## I. Problem

- **Reversal Curse:** autoregressive LMs trained on forward facts  $A \rightarrow B$  fail on the reversed queries  $B \rightarrow A$ .
- Prior solutions often require reversal training data, modified model architecture or modified training objectives.
- **Question:** Can we break the reversal curse without changing the model loss, architecture, or adding reversal training data?

Forward Training



Alice's husband is Bob.

Reversal Query (fails)



Who is Bob's wife?

## II. Preliminaries

**Dataset:**

- Forward:  $\mathcal{D}_{r_+} = \{[a_i, r_+] \rightarrow b_i : i \in [N]\}$
- Reversal:  $\mathcal{D}_{r_-} = \{[b_i, r_-] \rightarrow a_i : i \in [N]\}$
- Identity Bridge:  $\mathcal{D}_{\text{idn}} = \{[a_i, r_{\text{id}}] \rightarrow a_i : i \in [N]\} \cup \{[b_i, r_{\text{id}}] \rightarrow b_i : i \in [N]\}$

**One Layer Transformer Model:**

$$\text{TF}_{\theta}(x_{1:T}) := W_O W_V^T X \text{softmax}(X^T W_{KQ} x_T).$$

Where  $\theta = (W_O, W_V)$ .

Output logits and probability of token  $y$ :

$$\text{TF}_{\theta}(x_{1:T}; y) := z_y^T \text{TF}_{\theta}(x_{1:T}), \quad p_{\theta}(y | x_{1:T}) = \frac{\exp(\text{TF}_{\theta}(x_{1:T}; y))}{\sum_{v' \in V} \exp(\text{TF}_{\theta}(x_{1:T}; v'))}.$$

Cross entropy loss:

$$L_D(\theta) = \mathbb{E}_{[x_{1:T}, y] \sim D} [-\log p_{\theta}(y | x_{1:T})].$$

Logits difference:

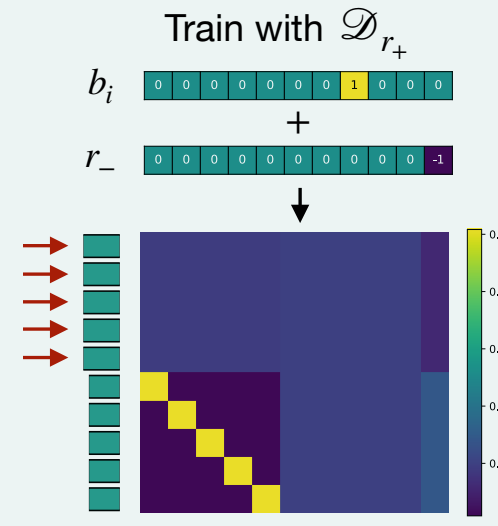
$$h_{[b_i, r_-], a_j}(\theta) = \text{TF}_{\theta}([b_i, r_-]; a_i) - \text{TF}_{\theta}([b_i, r_-]; a_j)$$

## III. Theory

**Theorem 3.3 (informal) [Reversal curse happens when trained with  $\mathcal{D}_{r_+}$ ]:** Under suitable assumptions, if the one-layer transformer is trained with  $\mathcal{D}_{r_+}$ , then the limiting direction of the weight  $W_O W_V^T$  satisfies:  $\forall i \in [N]$ ,

$$h_{[b_i, r_-], a_j}(W_O W_V^T) = 0, \forall i \neq j.$$

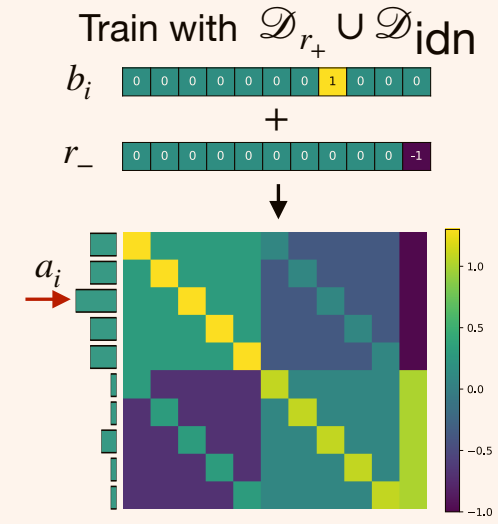
Thus, the model **can not** generalize to the reversal relation.



**Theorem 3.4 (informal) [Break the reversal curse with the identity bridge]:** Under suitable assumptions, if the one-layer transformer is trained with  $\mathcal{D}_{r_+} \cup \mathcal{D}_{\text{idn}}$ , then the limiting direction of the weight  $W_O W_V^T$  satisfies:  $\forall i \in [N]$ ,

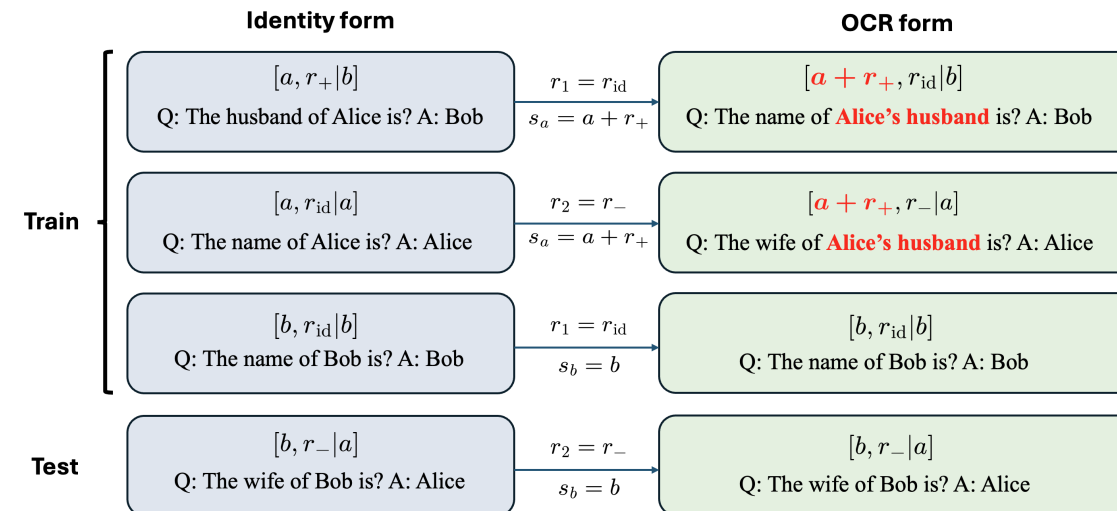
$$h_{[b_i, r_-], a_j}(W_O W_V^T) > 0, \forall i \neq j.$$

Thus, the model **can** generalize to the reversal relation.



**Why the identity bridge works?**

Identity bridge regularized reversal task is equivalent to an **Out-of-Context Reasoning (OCR)** task.



## IV. Experiment

**Model:** Llama-3.2-1B-Instruct **Task:** Husband-Wife and Parent-Child

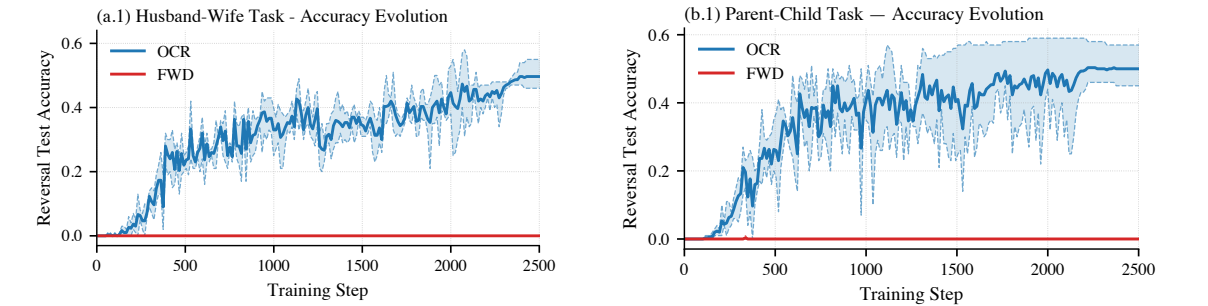
**Dataset:** take Husband-Wife task as an example

- (1)  $[a + r_+, r_{\text{id}} | b]$ : Q: The name of Alice's husband is? A: Bob.
- (2)  $[a + r_+, r_- | a]$ : Q: The wife of Alice's husband is? A: Alice.
- (3)  $[b, r_{\text{id}} | b]$ : Q: The name of Bob is? A: Bob.
- (4)  $[a, r_{\text{id}} | a]$ : Q: The name of Alice is? A: Alice.

FWD dataset: (1); OCR dataset: (1)  $\times$  k + (2) + (3); IDN dataset: (1) + (3) + (4)

Test set  $[b, r_- | a]$ : Q: The wife of Bob is? A: Alice.

(a). Experiment results on the two reversal tasks

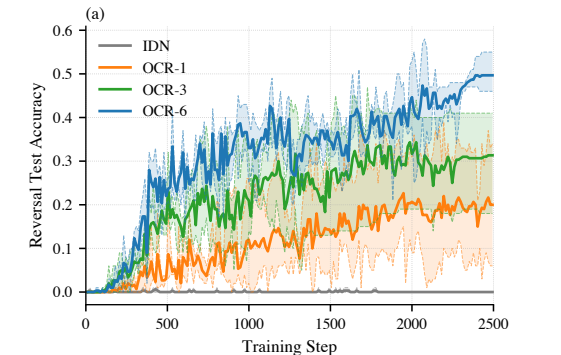


Forward facts only training suffers from reversal curse (**0%** accuracy).  
Identity bridge regularization breaks reversal curse (over **50%** accuracy).

(b). Necessity of the OCR form

The identity form (IDN) regularization (3) + (4) **can not** break the reversal curse.

Reversal accuracy improves with more repeated forward relation data (1).



(c). Ablation on the entity token length

- Number name: 1-token (e.g., 17)
- Normal name: 2-token (e.g., Sophia)
- Long name: 3-token (e.g., Catalina)

Entities with fewer tokens can be learned better in the reversal task

